

Muhammad Zohaib Iqbal

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EDUCATION AND TRAINING

Masters of Philosophy in Molecular Biology

National Centre for Excellence in Molecular Biology, University of the Punjab, Lahore [25 Sep 2019 – 3 Sep 2021]

Cumulative Grade Point Average: 3.44/4

Master's Project and Master's Thesis

Stem cell Lab, National Centre for Excellence in Molecular Biology (CEMB) [1 Aug 2020 – 1 Aug 2021]

"Caffeic preconditioned human adipose-derived mesenchymal stem cells co-transplanted with platelet-rich plasma improve wound healing in diabetic rats"

Skills:

1. Isolation, characterization, and preconditioning of adipose-derived mesenchymal stem cells against hypoxic stress *in-vitro*.
2. Transplantation of caffeic acid preconditioned human adipose-derived mesenchymal stem cells with platelet-rich plasma *in-vivo* with streptozotocin-induced diabetic models.

Bachelor of Science in Biotechnology

Forman Christian College University, Lahore [28 Jan 2014 – 5 Jul 2019]

Cumulative Grade Point Average : 3.03/4

WORK EXPERIENCE

Research Associate

Centre for Regenerative Medicine & Stem Cell Research, Aga Khan University Hospital Karachi [15 Nov 2021 – Current]

Working on a project titled **"Developing injectable CRISPR\Cas9 system for β -thalassemia and sickle cell disease (SCD)"**.

- To optimize *in-vitro* and *in-vivo* models of β -thalassemia
- To establish cell lines from primary cells derived from β -thalassemia patients

Research Associate

COMSATS University, Lahore [10 Aug 2021 – 10 Nov 2021]

Worked on a collaborative project titled "**Development of drug-loaded nano-composite scaffolds for skin tissue engineering applications**".

- To test the regenerative potential of different scaffolds *in-vivo* by using rat models
- To create excisional wound rat models
- To analyze the gene expression data from different stages of healing skin tissue using RT-PCR
- To assess wound healing by wound closure and re-epithelialization
- To perform immunohistochemistry on skin biopsies to analyze skin architecture

Research Associate

Stem Cell Lab, National Centre for Excellence in Molecular Biology Lahore [1 Aug 2020 – 1 Aug 2021]

- collecting and processing adipose tissue to isolate stem cells
- maintenance of animal facility

Research Associate

Molecular Virology Lab, National Centre for Excellence in Molecular Biology (CEMB) [1 Oct 2019 – 14 Mar 2020]

- collecting and storing blood samples for research
- isolating DNA from blood sample
- actively assisting in project designing
- designing and leading a project titled "**Gene expression study of drug-resistant *Salmonella typhi***"

University Teaching Assistant

Forman Christian College University (FCCU) Lahore, Pakistan [1 Jan 2018 – 28 Jun 2019]

- planning quizzes, assignment collection, correction, and presentations
- assisting with clerical tasks
- participating in student evaluation throughout the semester

Science Teacher

Yasin Model School Burewala [27 Dec 2016 – 15 Dec 2017]

- Teaching of biology and chemistry subjects to ninth and tenth grade students.

Intern

National Center of Excellence in Molecular Biology (CEMB), University of the Punjab [17 Jul 2016 – 29 Aug 2016]

- Isolation and maintenance of bacterial cultures
- DNA isolation from fungal and plant cultures
- Bacterial and plant transformation
- PCR and gel-electrophoresis

Science Teacher

Yasin Model School Burewala [3 Jan 2011 – 7 Jul 2011]

- Teaching of different scientific subjects to sixth and seventh-grade students.

RESEARCH ARTICLE

Design and development of thyroxine/heparin releasing affordable cotton dressings to treat chronic wounds

Waris TS, Andleeb A, Bazzar M, Moradi AR, **Iqbal MZ**, Yar M. Journal of Tissue Engineering and Regenerative Medicine, 2022 (Under Review)

CONFERENCES AND SEMINARS

NGS Technology (Illumina) and its Applications

[Centre for Applied Molecular Biology (CAMB), Lahore Pakistan, 14 Oct 2021]

International Symposium on "STEM CELLS AND REGENERATIVE MEDICINE"

[Centre of Excellence in Molecular Biology, University of the Punjab, 15 Nov 2019]

- Organized and participated in the event
- Lead a team of volunteers who managed the event

International Symposium on "BRAIN NEUROGENETICS AND REGENERATIVE MEDICINE"

[Centre of Excellence in Molecular Biology, University of the Punjab, 3 Oct 2019]

Invention to Innovation Summit (Buy and Sell Technology)

[National Centre of Excellence in Molecular biology, University of the Punjab, Lahore, 2 Apr 2019 – 3 Apr 2019]

POSTER PRESENTATION

Alternative ways to produce heat stable enzymes

[18 May 2019]

At the **2nd Forman Christian College innovation competition (FCC-Innovation)** in Forman Christian College University, Lahore

SKILLS

Research Skills

- Stem cell isolation
- Stem cell culturing
- Live cell imaging
- PKH26 labelling
- Cell viability assays
- Animal handling
- Tissue processing
- PCR
- Agarose Gel electrophoresis
- RT-PCR
- Hematoxylin/Eosin staining
- Fluorescence microscopy
- Cytogenetics

Bioinformatics

Databases

- NCBI
- UCSC Genome Browser
- Expasy: SIB

Tools

- Primer Designing
- ImageJ
- BLAST

ONLINE COURSES

Coursera online certificates

1. How to write and publish a scientific paper (project-centered course)
2. Understanding medical research
3. Introduction to the biology of cancer
4. Introduction to breast cancer
5. Understanding cancer metastasis
6. Introduction to genomic technologies
7. Science of stem cells
8. Teaching science at university

WEBINARS

Labroots

1. CRISPR 2021
2. Troubleshooting cell-based assays
3. Automating culture and high-content imaging of 3D organoids for *in-vitro* assessment of compound effects
4. Delivering the cells that matter : accelerating genome engineering workflows through efficient single cell cloning of human iPSCs

NEWSPAPER ARTICLE

Coronavirus Fidelity: Carry Logical Fears

Mustafa M, & Iqbal MZ. Coronavirus fidelity: carry logical fears. Technology Times, 2020

<https://www.technologytimes.pk/2020/02/09/coronavirus-fidelity-carry-logical-fears/>

VOLUNTEERING

Lahore Science Mela 2019

[Ali Educational institute, Organized by Khwarzmi Science Society]

Lahore Science Mela (LSM) is a Cambridge Science Festival for scientists, artists, science connoisseurs, technology enthusiasts, photographers, inventors, entrepreneurs and the larger general public.

<https://khwarizmi.org/lahore-science-mela/>

Festival of Life

[Shaukat Khanum Memorial Cancer Hospital and Research Centre, 28 Feb 2016]

Leading a team of volunteers to organize the Festival of Life. It is an event for fundraising for Shaukat Khanum Cancer Hospital Pakistan.

<https://www.hipinpakistan.com/news/1149161>